
MA 608: Computational Fluid Dynamics

Course Code	MA 608
Course Name	Computational Fluid Dynamics
L-T-P-C	3-0-0-3
Prerequisites	Basic knowledge in Fluid Mechanics
Intended for	MS/ Ph.D. and UG
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Introduction**

Historical Perspective, Comparisons of experimental, Theoretical and Numerical approaches. Different numerical Approaches. [4 lectures]

- **Governing Equations**

Classification of Partial Differential Equations, Physical Classification, Mathematical Classification, Well-posed problems, Navier-Stokes System of equations. [6 lectures]

- **Finite Difference Methods**

Derivation of Finite Difference Equations, Simple Methods, General Methods, Multidimensional Formulas, Accuracy of Finite Difference solutions. [6 lectures]

- **Solution Methods of Finite Difference Equations**

Elliptic Equations, Parabolic Equations, Hyperbolic Equation, Example Problems, Stability, Convergence and Consistency of the Solution methods. [14 lectures]

- **Application of Finite Difference Methods to the Equations of Fluid Mechanics**

Numerical Methods for Inviscid Flow Equations, Numerical Methods for Boundary-Layer Type Equations. [9 lectures]

- **Introduction to Finite Volume Methods**

Basic Formulations, SIMPLE algorithm. [4 lectures]

Text and Reference Books

1. D. A. Anderson, J. C. Tannehill, and R. H. Pletcher, **Computational Fluid Mechanics and Heat Transfer**, 2nd Edition, Taylor & Francis, 1997.
2. J. D. Anderson Jr, **Computational Fluid Dynamics**, McGraw-Hill International Edition, 1995.
3. S. V. Patankar, **Numerical Heat Transfer and Fluid Flow**, Hemisphere, 2000.
4. T. J. Chung, **Computational Fluid Dynamics**, 2nd Edition, Cambridge University Press, 2010. item P. Niyogi, S. K. Chakrabartty, M. K. Laha, **Introduction to Computational Fluid Dynamics**, Pearson Publications, 2011.